LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Procedure of turning, facing, thread cuttings, grooving, knurling and tapering in lathe machine
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Procedure of turning, facing, thread cuttings, grooving, knurling and tapering in lathe machine.
5. Specific Objectives	31.1.1.1 Trainee will be able to perform various general engineering works

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Lathe machine operation: Turning, Facing, Threading, Grooving, Knurling, Tapering; Safety precautions; Instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a lathe machine?
(b) List the six operations covered in this lesson.
(c) What is the first safety step before starting a lathe?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Turning operation procedure • Facing operation procedure • Thread cutting procedure • Grooving operation procedure • Knurling operation procedure	L & Ds	PPT, IB	15 min
(b)	• Tool selection for each operation • Speed and feed settings • Workpiece clamping techniques • Cutting fluid application • Measurement and inspection methods	L & Ds	PPT, PDF	20 min
(c)	• Common defects and causes • Troubleshooting techniques • Quality control standards • Maintenance of lathe tools • Safety during operation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between grooving and knurling.
(b) Describe the procedure for cutting threads on a lathe.
(c) List three safety precautions specific to threading operations.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between turning and facing operations, including tool selection and feed rates, to be submitted before Lesson Plan 03.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	The procedure of Gear cutting in Milling machine	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min